

Master's Seminar: Artificial Intelligence - Visual Computing

3D Gaussian Splatting

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1 Introduction and Motivation

2 3D Gaussian Data Structure

3 Training Procedure

- Point Cloud Initialization
- Forward Pass
- Backward Pass

4 Novel-View Synthesis

5 Video and Discussion

Core Concepts

3D Scene Creation

Novel-View Synthesis

Radiance Field

Splatting

Definitions

Creates 3D environments such as individual objects or outdoor spaces from an overlapping set of 2D input images.

Core Concepts

3D Scene Creation

Novel-View Synthesis

Radiance Field

Splatting

Definitions

Synthesizes new 2D imagery from the 3D scene using new camera positions.

Real-time capability requires at least 24–30 frames per second (FPS) to appear smooth enough for the human eye.

Core Concepts

3D Scene Creation

Novel-View Synthesis

Radiance Field

Splatting

Definitions

A function mapping coordinates and viewing angles to color and opacity at the given coordinates.

Implicit fields are continuous; explicit fields use a finite set of points.

Core Concepts

3D Scene Creation

Novel-View Synthesis

Radiance Field

Splatting

Definitions

The process of moving all elements inside the view frustum onto the view plane preserving order, adding to aggregated color and opacity of each pixel.

Core Concepts

3D Scene Creation

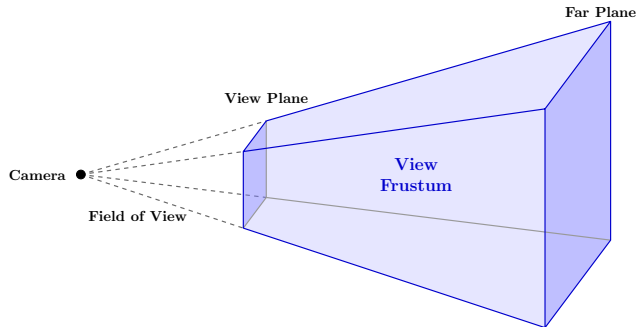
Novel-View Synthesis

Radiance Field

Splatting

Definitions

“Pyramid of Vision”



Characteristics & Limits

Implicit Representation

Volumetric Ray Marching

Training Bottlenecks

“Black Box” Inflexibility

Quality vs. Speed

Details

NeRFs learn an implicit radiance field representation, meaning the full 3D scene is encoded entirely within the neural network's weights.

Characteristics & Limits

Implicit Representation

Volumetric Ray Marching

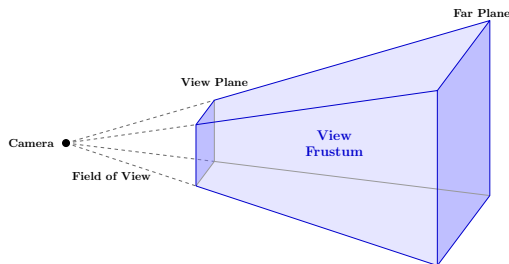
Training Bottlenecks

“Black Box” Inflexibility

Quality vs. Speed

Details

To synthesize novel views, NeRFs transmit a ray for every pixel, resulting in a very high number of queries to the neural network for discretized points along the ray's path.



Characteristics & Limits

Implicit Representation

Volumetric Ray Marching

Training Bottlenecks

“Black Box” Inflexibility

Quality vs. Speed

Details

Ray marching and splatting are opposite ways to determine a pixel color value.

Characteristics & Limits

Implicit Representation

Volumetric Ray Marching

Training Bottlenecks

“Black Box” Inflexibility

Quality vs. Speed

Details

Training NeRFs to reach photorealistic scene quality can take several days on professional high-end GPU hardware.

Ray marching during synthesis is highly time-consuming.

Characteristics & Limits

Implicit Representation

Volumetric Ray Marching

Training Bottlenecks

“Black Box” Inflexibility

Quality vs. Speed

Details

Because all scene details are hidden inside the neural network's weights, it is not easily possible to edit or amend a reconstructed scene.

Characteristics & Limits

Implicit Representation

Volumetric Ray Marching

Training Bottlenecks

“Black Box” Inflexibility

Quality vs. Speed

Details

NeRFs force a trade-off: state-of-the-art reconstruction quality requires long training and synthesis times.

Abandoning the neural network for explicit representations increases speed but inherently reduces image quality.

Core Innovations

Origin & Architecture

The Base Primitive

Fully Differentiable

Direct Editability

Hardware Optimization

Details

Introduced by Kerbl et al. at INRIA in France in late 2023, 3D Gaussian Splatting (3DGS) combines existing concepts to create a learnable explicit radiance field representation.

The publication had a very high impact in the field with over 10,000 citations in three years.

Core Innovations

Origin & Architecture

The Base Primitive

Fully Differentiable

Direct Editability

Hardware Optimization

Details

A 3D Gaussian is an ellipsoid “fuzzy blob” that loses opacity towards its edges and can take shapes ranging from perfect spheres to elongated needles or flat shapes.

Using them as base primitives bypasses previous issues with explicit fields and allows for arbitrary positional precision.

Core Innovations

Origin & Architecture

The Base Primitive

Fully Differentiable

Direct Editability

Hardware Optimization

Details

By cleverly decomposing the covariance matrix, 3D Gaussians become differentiable and eligible for standard gradient descent backpropagation.

Most importantly, this means 3DGS operates completely without neural networks.

Core Innovations

Origin & Architecture

The Base Primitive

Fully Differentiable

Direct Editability

Hardware Optimization

Details

Unlike the neural network “black box” of NeRFs, 3D Gaussians provide direct editability of the scene.

This makes scene modifications, such as additions or lighting changes, highly accessible for downstream applications.

Core Innovations

Origin & Architecture

The Base Primitive

Fully Differentiable

Direct Editability

Hardware Optimization

Details

Kerbl et al. mapped their architecture exhaustively onto modern GPU hardware.

The centerpiece is the tile-based rasterizer, which allows real-time novel-view synthesis at 1080p resolution while matching or improving upon state-of-the-art NeRF quality.

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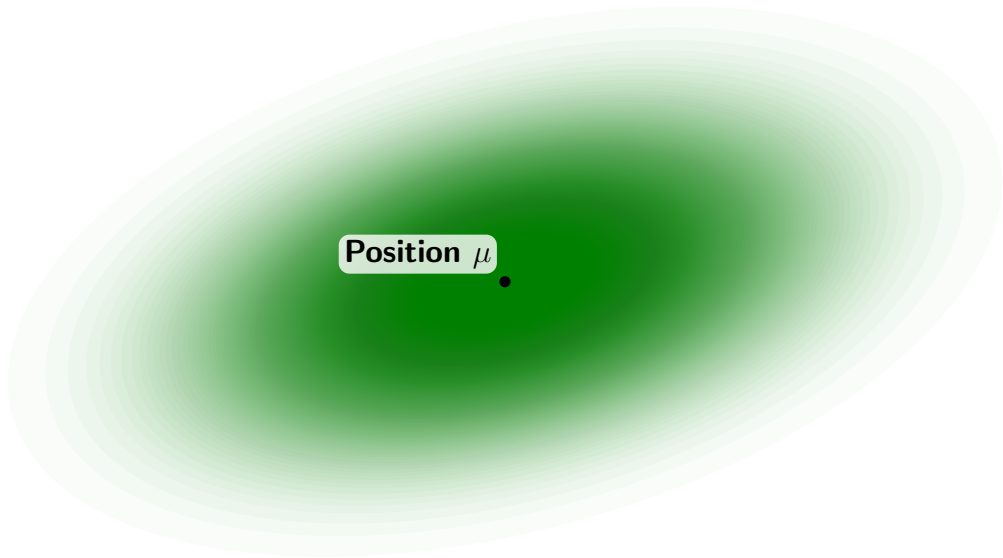
3 Training Procedure

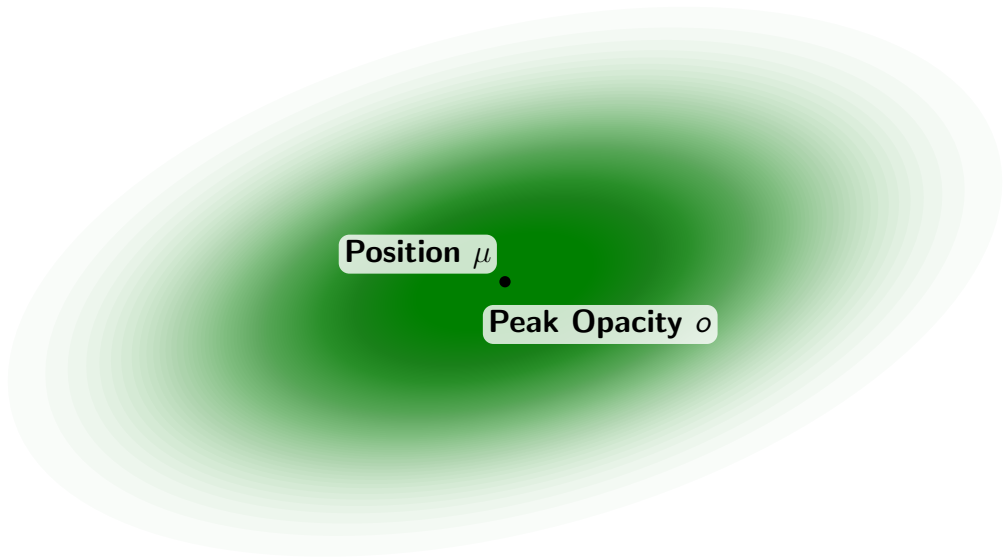
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- Forward Pass
- Backward Pass

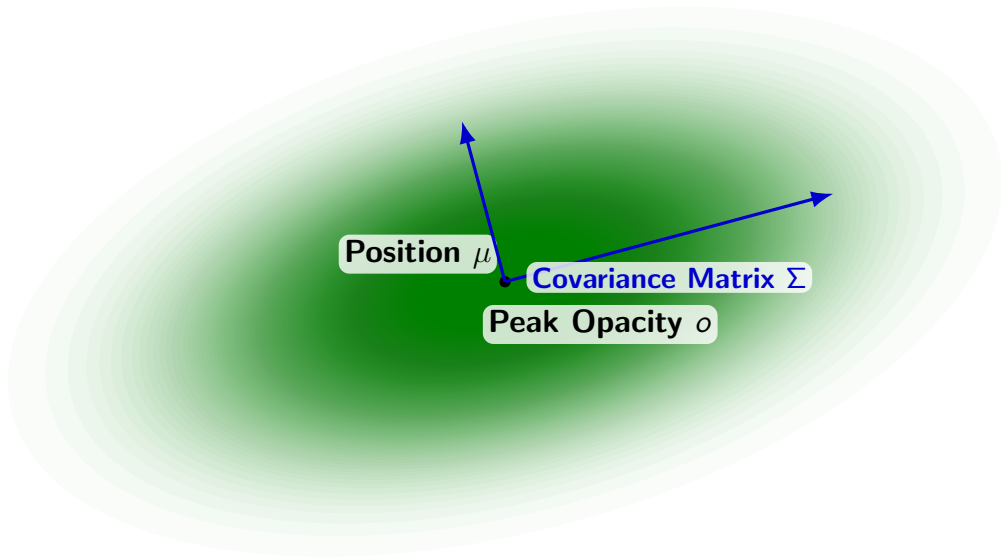
4 Novel-View Synthesis

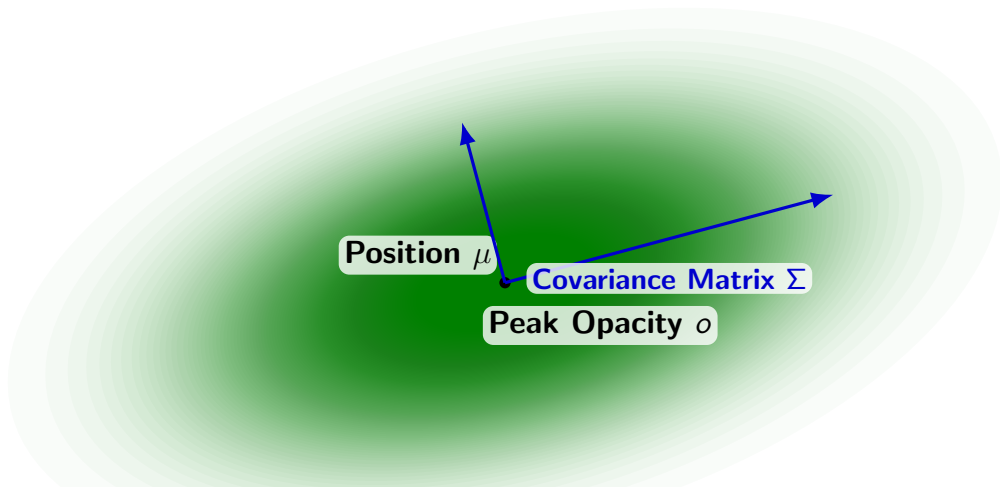
5 Video and Discussion



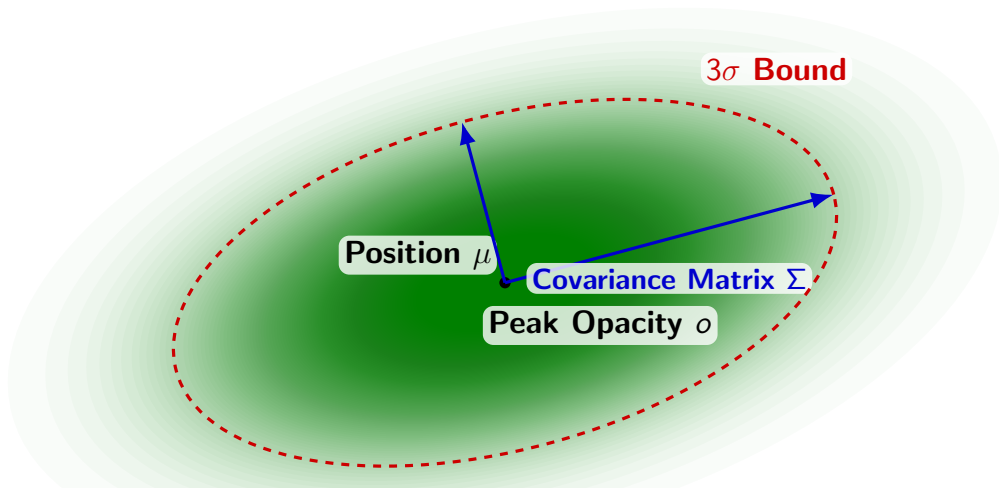








Multivariate Normal Distribution: $\alpha(p) = o \cdot \exp\left(-\frac{1}{2}(p - \mu)^T \Sigma^{-1}(p - \mu)\right)$



$$\text{Multivariate Normal Distribution: } \alpha(p) = o \cdot \exp\left(-\frac{1}{2}(p - \mu)^T \Sigma^{-1}(p - \mu)\right)$$

Mathematical Setup

The Validity Problem

Core Decomposition

Scaling Values

Rotational Quaternion

Guaranteed Validity

Details

To work with gradient descent, the covariance matrix Σ must be differentiable and remain positive semi-definite (eigenvalues ≥ 0).

Running a loop during backpropagation to find valid values is computationally too costly.

Mathematical Setup

The Validity Problem

Core Decomposition

Scaling Values

Rotational Quaternion

Guaranteed Validity

Details

Kerbl et al. bypassed this by decomposing Σ into a scaling matrix S and rotational matrix R mirroring standard eigendecomposition ($\Sigma = Q\Delta Q^T$):

$$\Sigma = RSS^T R^T \text{ with } \Delta = S^2, Q = R$$

Mathematical Setup

The Validity Problem

Core Decomposition

Scaling Values

Rotational Quaternion

Guaranteed Validity

Details

The diagonal scaling matrix S is further decomposed into three distinct scaling values (s_1, s_2, s_3) . These represent the roots of the eigenvalues of Σ .

Mathematical Setup

The Validity Problem

Core Decomposition

Scaling Values

Rotational Quaternion

Guaranteed Validity

Details

The rotational matrix R is decomposed into a four-value quaternion (w, x, y, z) :

$$R = \begin{bmatrix} 1 - 2(y^2 + z^2) & 2(xy - wz) & 2(xz + wy) \\ 2(xy + wz) & 1 - 2(x^2 + z^2) & 2(yz - wx) \\ 2(xz - wy) & 2(yz + wx) & 1 - 2(x^2 + y^2) \end{bmatrix}$$

Mathematical Setup

The Validity Problem

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Rotational Quaternion

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Details

3DGS only stores and optimizes these 7 specific values for each 3D Gaussian.

Because an eigendecomposition is always positive semi-definite, constructing a new covariance matrix from these components is guaranteed to be geometrically valid.

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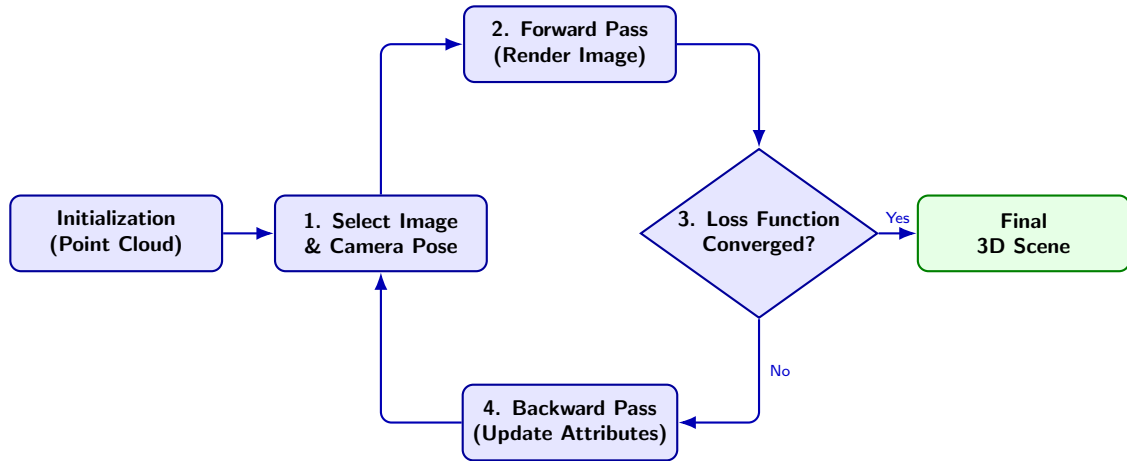
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The Training Cycle



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Initialization Steps

Structure from Motion

Camera Matrix

Primitive Initialization

Details

The 3DGS algorithm requires initial geometry, which is commonly derived from a set of input images.

Because camera poses are usually unknown, Structure from Motion (SfM) software like COLMAP is used to solve a geometric optimization problem, providing an accurate 3D point cloud.

Initialization Steps

Structure from Motion

Camera Matrix

Primitive Initialization

Details

As another result from SfM, each input image is assigned a 3×3 camera matrix W_{rot} which contains the camera's orientation in world space.

Initialization Steps

Structure from Motion

Camera Matrix

Primitive Initialization

Details

The SfM points act as the origins for the initial 3D Gaussians.

- ▶ They are initialized as perfect spheres.
- ▶ Radii are deduced from their three nearest neighbors.
- ▶ Opacity is set to 0.1 and color to any RGB value.

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- ▶ The GPU is running one GPU thread per 3D Gaussian.

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Phase 1: Preparing Constituent Attributes

- ▶ Raw values optimized during training are converted from latent space back into physical space.
- ▶ Opacity is bounded to $(0, 1)$ using a sigmoid function, scale uses the exponential function, and the rotational quaternion is normalized.

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Phase 2: Reconstructing the 3D Covariance Matrix

- ▶ The algorithm mathematically rebuilds the 3D geometric shape.
- ▶ It combines the scaling matrix S and rotational matrix R to form the full 3D covariance matrix:

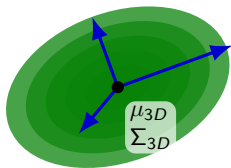
$$\Sigma_{3D} = RSS^T R^T$$

Phase 3: Projection from 3D to 2D

- ▶ Converts 3D Gaussians (ellipsoids) in world space into flat 2D Gaussians (ellipses) in image space.
- ▶ Employs a linear affine approximation (the Jacobian matrix J) of a standard non-linear pinhole projection.
- ▶ The 3D matrix is rotated into view space via W_{rot} and then projected:

$$\Sigma_{2D} = JW_{rot}\Sigma_{3D}W_{rot}^TJ^T$$

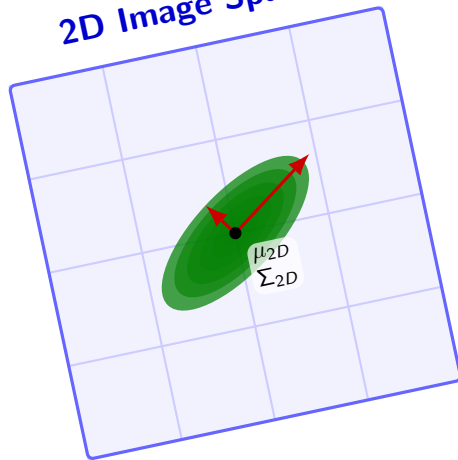
3D World Space



Projection
(Jacobian J)



2D Image Space



Phase 4: Preparing Radix Sort

- ▶ The target output image is divided into a grid of 16×16 pixel tiles.
- ▶ The algorithm calculates a 2D bounding box to map exactly which tiles a given 2D Gaussian covers.
- ▶ Assigns each 2D Gaussian to a tile. If overlapping, assigns to all affected tiles.

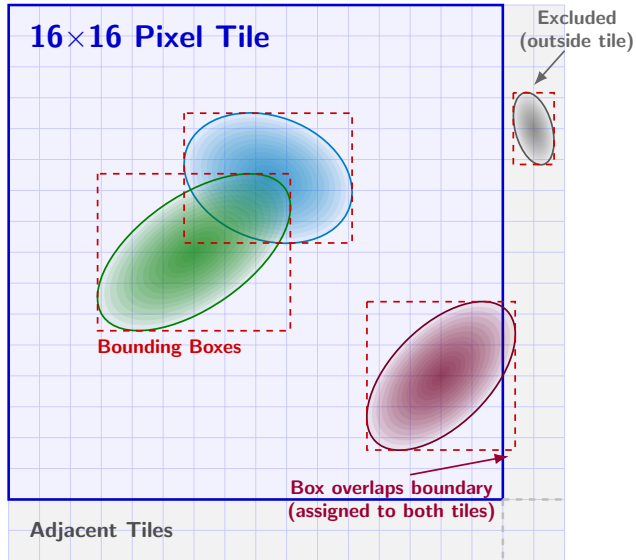
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Phase 5: Global Radix Sort

- ▶ The GPU waits until all per-Gaussian computations are finished.
- ▶ A highly optimized Radix Sort is executed, organizing the 2D Gaussians for each tile strictly by depth (closest to farthest).

Forward Pass: Phases 4 & 5

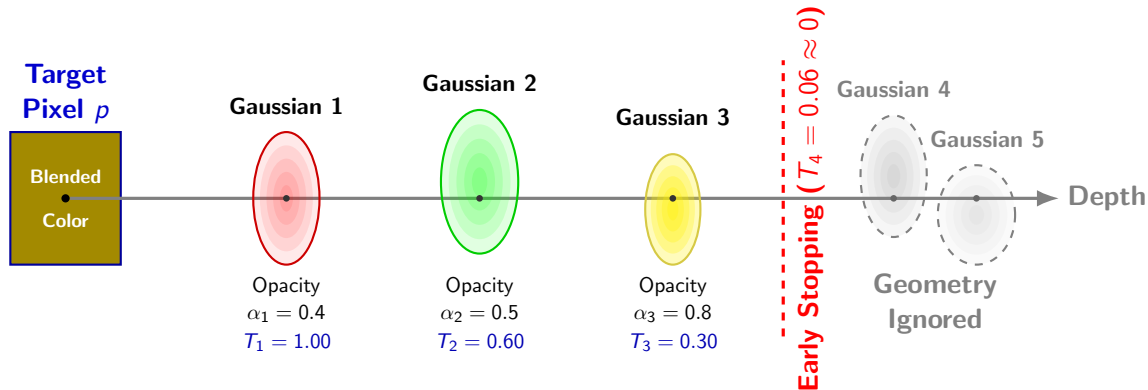


Phase 6: Tile-Based Rasterization (Splatting)

- ▶ Operating natively in 2D image space, the GPU assigns one processing thread per pixel.
- ▶ **Alpha Compositing:** Pixels accumulate color by blending the sorted list of overlapping 2D Gaussians front-to-back:

$$C_p = \sum_{i=1}^N c_i \alpha_i T_i, \quad \text{with} \quad T_i = \prod_{j=1}^{i-1} (1 - \alpha_j) \quad \text{the transmittance}$$

- ▶ **Early Stopping:** Threads stop calculating exactly when transmittance (T_i) drops near zero, ignoring hidden geometry for massive speed-ups.



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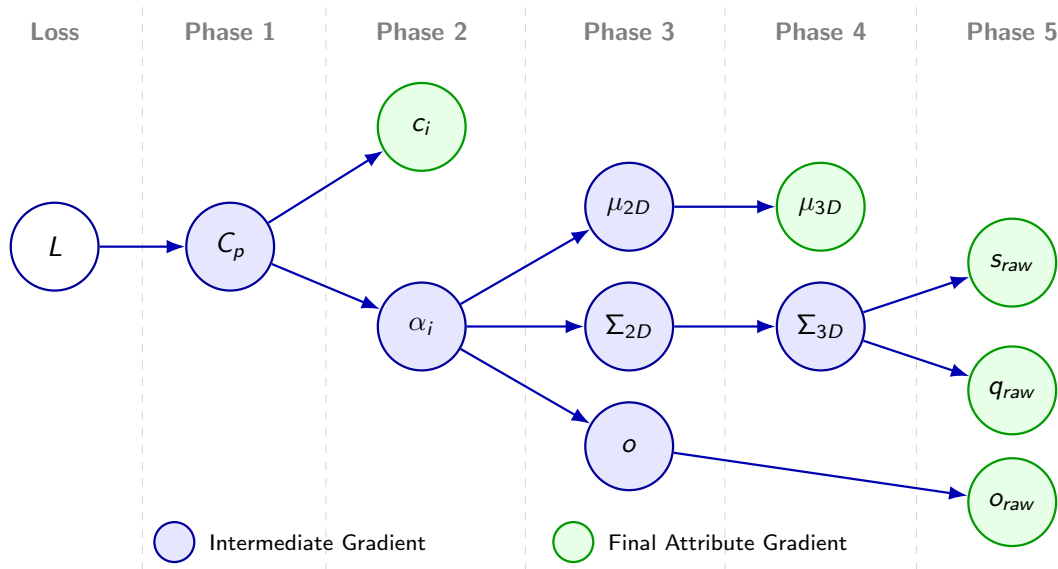
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Backward Pass: Gradient Flow Overview



The Loss Function

- ▶ Training compares the rendered image with the original input image using a weighted combination of $\mathcal{L}_1$ loss and D-SSIM.
- ▶ $\mathcal{L}_1$ calculates the absolute pixel-wise error, while D-SSIM acts as a structural regularizer:

$$L = (1 - \lambda) \cdot \mathcal{L}_1 + \lambda \cdot \mathcal{L}_{D-SSIM}, \quad \text{with } \lambda = 0.2$$

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Phase 1: Loss to Pixel Color Correction

- ▶ The gradient calculates the required color correction per pixel (C_p).
- ▶ A positive gradient indicates an overestimation of color, negative indicates underestimation, and the magnitude dictates the degree of adjustment required:

$$\frac{\partial L}{\partial C_p} = \frac{1 - \lambda}{N} \cdot \text{sign}(C_p - O_p) + \lambda \frac{\partial \mathcal{L}_{D-SSIM}}{\partial C_p}$$

Phase 2: Pixel Color Correction to 2D Splat

- ▶ The algorithm works alpha compositing backwards, flowing the pixel gradient into the evaluated opacity (α_i) and color (c_i) of each 2D Gaussian in the tile.

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- ▶ The algorithm works alpha compositing backwards, flowing the pixel gradient into the evaluated opacity (α_i) and color (c_i) of each 2D Gaussian in the tile.
- ▶ **Color Gradient:** Uses the chain rule to update the Gaussian's color based on its contribution (transmittance T_i and opacity α_i):

$$\text{with } C_p = \sum_{i=1}^N c_i \alpha_i T_i : \quad \frac{\partial L}{\partial c_i} = \frac{\partial L}{\partial C_p} \cdot \frac{\partial C_p}{\partial c_i} = \frac{\partial L}{\partial C_p} \cdot (\alpha_i T_i)$$

Phase 2: Pixel Color Correction to 2D Splat

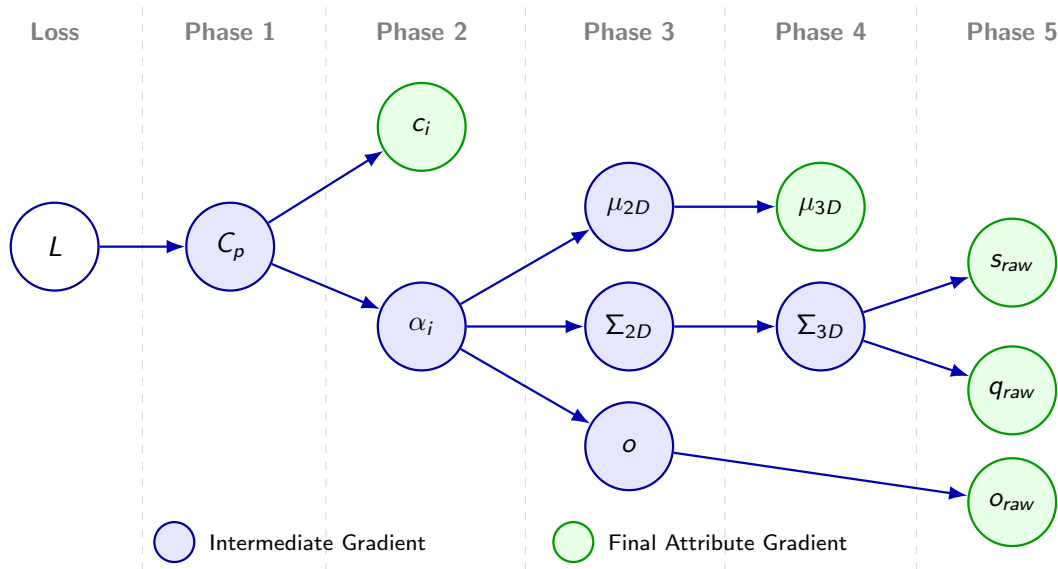
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- ▶ **Opacity Gradient:** Increases evaluated opacity if the Gaussian's color (c_i) reduces the pixel error more effectively than the accumulated background color behind it (C_{behind}):

$$\frac{\partial L}{\partial \alpha_i} = \frac{\partial L}{\partial C_p} \cdot \frac{\partial C_p}{\partial \alpha_i} = \frac{\partial L}{\partial C_p} \cdot T_i (c_i - C_{behind})$$

Backward Pass: Gradient Flow Overview



Phase 3: 2D Splat to 2D Geometry

- ▶ The opacity gradients are used to adjust the physical properties of the 2D Gaussians.
- ▶ This mathematically pulls the center position of the 2D Gaussian towards the pixel and stretches or contracts its 2D shape to minimize the error.

Phase 3: 2D Splat to 2D Geometry

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Phase 4: 2D Geometry to 3D Geometry

- ▶ The GPU model changes from pixel-based to running per-Gaussian calculations.
- ▶ The 2D gradients are unprojected back into 3D space, translating the 2D adjustments into 3D center movements and 3D volume changes.

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Phase 5: 3D Geometry to 3D Gaussian Attributes

- ▶ The gradients flow backward through the 3D covariance matrix assembly.
- ▶ Adjustments are mapped directly into the raw, latent space variables for scale, rotation, and base opacity.

Adaptive Density Control

- ▶ Steps in every 100 iterations to counter over-reconstruction (covering too much geometry) and under-reconstruction (too few Gaussians).
- ▶ **Splitting:** Over-reconstructing Gaussians are split into two smaller ones.
- ▶ **Cloning:** Under-reconstructing Gaussians are cloned.

Adaptive Density Control

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Pruning & Opacity Reset

- ▶ 3D Gaussians with extremely low opacity or unbound scale are routinely deleted.
- ▶ Every 3,000 iterations, the base opacity is reset to almost 0 for all Gaussians. This forces unimportant Gaussians to fade out completely and be pruned.

Adaptive Density Control

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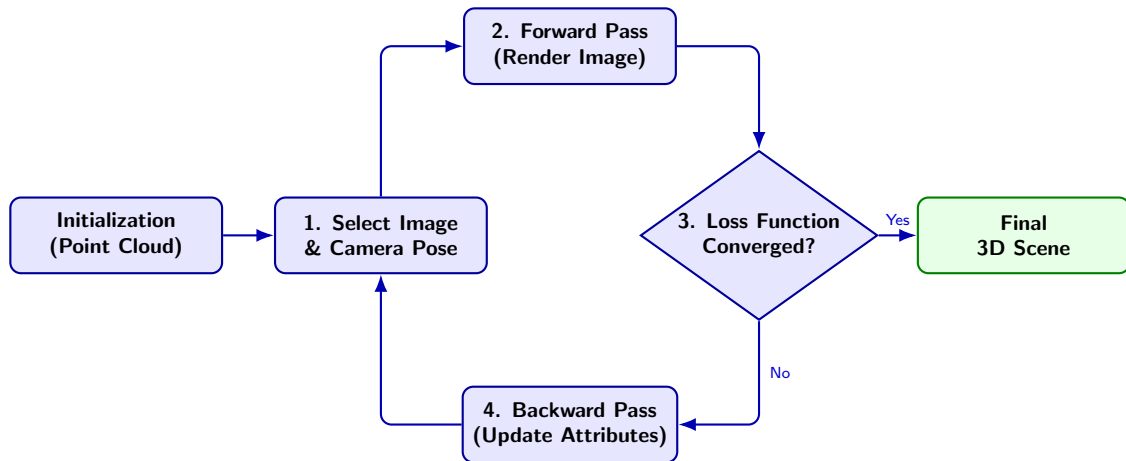
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Optimization Step

- ▶ Rather than standard gradient descent, 3DGS utilizes the Adam optimizer.
- ▶ Adam takes each parameter's history into account, smoothing out updates so that 3D Gaussians gently morph into their correct geometric shapes.

The Training Cycle



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Synthesis Steps

Pre-computation

Feed-Forward Execution

Disabled Components

Real-Time Performance

Details

Once the scene is fully trained, the algorithm performs Phase 2 of the forward pass one final time. This permanently calculates and stores the 3D covariance matrix for all 3D Gaussians.

Synthesis Steps

Pre-computation

Feed-Forward Execution

Disabled Components

Real-Time Performance

Details

From then on, generating new images is strictly feed-forward. The algorithm repeatedly executes only Forward Pass Phases 3 to 6 (projection, sorting, and rasterization) using the camera matrix of any desired novel view.

Synthesis Steps

Pre-computation

Feed-Forward Execution

Disabled Components

Real-Time Performance

Details

Because the scene geometry is finalized, the backward pass, Adaptive Density Control, and the Adam optimizer are completely disabled.

Synthesis Steps

Pre-computation

Feed-Forward Execution

Disabled Components

Real-Time Performance

Details

Freed from the training overhead and powered by the raw speed of the tile-based rasterizer, 3DGS quickly reaches real-time rendering levels.

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Training in Progress

<https://youtu.be/F9rgbvhSad8?t=550>



<https://youtu.be/yd40zUnr7eQ>



Real-Time Rendering

Example from the original paper's project website

<https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/content/videos/bicycle.mp4>



Looking forward to answering your questions.

6 Appendix

Phase 3: 2D Splat to 2D Geometry (Opacity & Position)

- ▶ We recall the simplified opacity distribution: $\alpha_i(p) = o \cdot e^E$, where E is the exponent derived from the pixel distance Δ and the inverse 2D covariance matrix.
- ▶ **Base Opacity (o):** The derivative scales directly with the distance from the center of the 2D Gaussian (e^E):

$$\frac{\partial L}{\partial o} = \frac{\partial L}{\partial \alpha_i} \cdot e^E$$

- ▶ **Position (μ_{2D}):** Let vector $v = \Sigma_{2D}^{-1} \Delta p$. This vector points directly from the 2D Gaussian's center to the pixel:

$$\frac{\partial L}{\partial \mu_{2D}} = \frac{\partial L}{\partial \alpha_i} \cdot \alpha_i(p) \cdot v$$

- ▶ If the evaluated opacity gradient is positive, the center is effectively pulled towards the pixel.

Phase 3: 2D Splat to 2D Geometry (Shape)

- ▶ **2D Covariance Matrix (Σ_{2D}):** We must calculate how the opacity at pixel p changes when the 2D shape of the Gaussian changes.
- ▶ This uses the same vector v from the position derivative, multiplied by its own transpose:

$$\frac{\partial L}{\partial \Sigma_{2D}} = \frac{\partial L}{\partial \alpha_i} \cdot \frac{1}{2} \alpha_i(p) \cdot (v \cdot v^T)$$

- ▶ Multiplying the vector by its transpose results in a matrix gradient.
- ▶ If the evaluated opacity gradient is positive, this matrix gradient causes the 2D Gaussian to stretch itself towards the pixel; otherwise, it causes it to contract.

Phase 4: 2D Geometry to 3D Geometry

- ▶ The GPU thread model changes to running per-Gaussian to unproject the 2D data back into 3D.
- ▶ This utilizes the multi-variable chain rule and the transposes of the forward projection matrices ($U = JW_{rot}$).
- ▶ **3D Position** (μ_{3D}): The gradient passes through view space (t) back to world space:

$$\frac{\partial L}{\partial \mu_{3D}} = W_{rot}^T J^T \frac{\partial L}{\partial \mu_{2D}} = U^T \left(\frac{\partial L}{\partial \mu_{2D}} \right)$$

- ▶ **3D Covariance** (Σ_{3D}): The final gradient unprojects cleanly using matrix U on both sides:

$$\frac{\partial L}{\partial \Sigma_{3D}} = U^T \left(\frac{\partial L}{\partial \Sigma_{2D}} \right) U$$

Phase 5: 3D Geometry to Matrix Components

- ▶ Gradients flow through the covariance matrix assembly ($\Sigma_{3D} = MM^T$ with $M = RS$) into scale (S) and rotation (R).
- ▶ First, the algorithm derives the intermediate matrix M :

$$\frac{\partial L}{\partial M} = 2 \left(\frac{\partial L}{\partial \Sigma_{3D}} \right) M$$

- ▶ Then, it splits this gradient into the separate components:

$$\frac{\partial L}{\partial S} = R^T \left(\frac{\partial L}{\partial M} \right) \quad \text{and} \quad \frac{\partial L}{\partial R} = \left(\frac{\partial L}{\partial M} \right) S$$

- ▶ The scale gradient ($\frac{\partial L}{\partial s}$) is extracted from the diagonal of $\frac{\partial L}{\partial S}$, while the quaternion gradient ($\frac{\partial L}{\partial q}$) aggregates the partial derivatives of R .

Phase 5: Backpropagating to Latent Attributes

- ▶ To close the loop, the gradients pass backwards through their respective activation functions to reach the raw, latent space variables.
- ▶ **Scale (Exponential):** Backpropagating through exp requires multiplying by s :

$$\frac{\partial L}{\partial s_{raw}} = \frac{\partial L}{\partial s} \cdot s$$

- ▶ **Rotation (Normalization):** Uses the Jacobian matrix ($J_{||}$) of the normalization function, with $J_{||} = \frac{I - qq^T}{||q_{raw}||}$:

$$\frac{\partial L}{\partial q_{raw}} = J_{||} \cdot \frac{\partial L}{\partial q}$$

- ▶ **Base Opacity (Sigmoid):** Backpropagating through the sigmoid function:

$$\frac{\partial L}{\partial o_{raw}} = \frac{\partial L}{\partial o} \cdot o(1 - o)$$